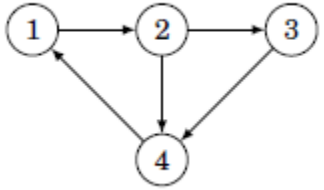
Graph'ların gösterimi

Komşuluk Gösterimi



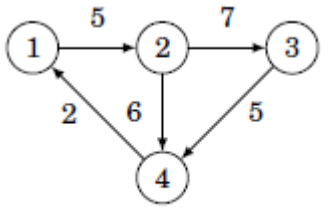

```
vector<int> adj[N];
```

```
adj[1].push_back(2);
adj[2].push_back(3);
adj[2].push_back(4);
adj[3].push_back(4);
adj[4].push_back(1);
```

Eğer graph yönlü değilse, her bir kenar çift yönlü olarak eklenerek gösterilir

Graph’ların gösterimi

Komşuluk Gösterimi (Weighted graph)

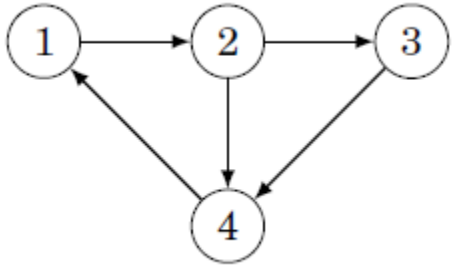



```
vector<pair<int,int>> adj[N];
```

```
adj[1].push_back({2,5});
adj[2].push_back({3,7});
adj[2].push_back({4,6});
adj[3].push_back({4,5});
adj[4].push_back({1,2});
```

Graph’ların gösterimi

Matris Gösterimi




```
int adj[N][N];
```

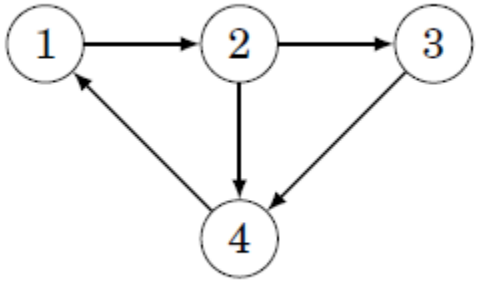
	1	2	3	4
1	0	1	0	0
2	0	0	1	1
3	0	0	0	1
4	1	0	0	0

Eğer a düğümünden b düğümüne bir kenar varsa adj[a][b] = 1 değilse 0’dır.

Weighted graph’lar için
adj[a][b] değeri kenarın değeridir.

Graph'ların gösterimi

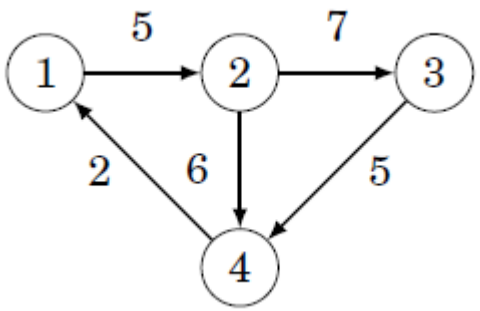
Kenar Gösterimi




```
vector<pair<int,int>> edges;
```

```
edges.push_back({1,2});
edges.push_back({2,3});
edges.push_back({2,4});
edges.push_back({3,4});
edges.push_back({4,1});
```

Weighted graph'lar için



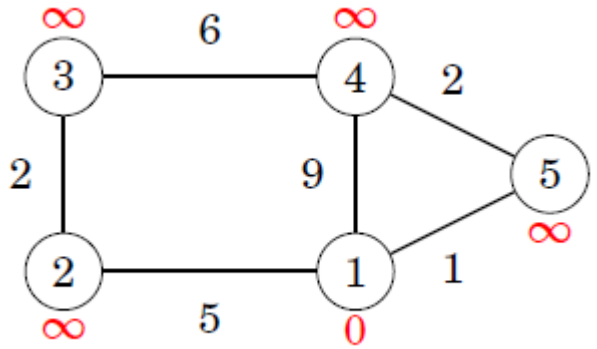

```
vector<tuple<int,int,int>> edges;
```

```
edges.push_back({1,2,5});
edges.push_back({2,3,7});
edges.push_back({2,4,6});
edges.push_back({3,4,5});
edges.push_back({4,1,2});
```

En kısa yol algoritmaları

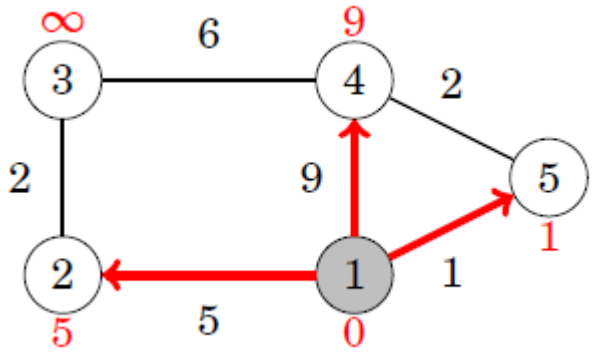
Bellman-Ford Algorithm O(nm)
SPFA Algorithm (Shortest Path Faster Algorithm) O(nm)
Dijkstra's Algorithm
Floyd Warshall Algorithm

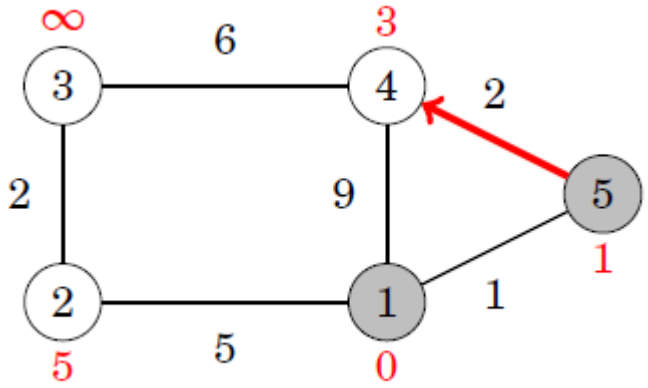
Dijkstra's negatif ağırlığı olan grafiklerde en kısa yolu garanti etmez

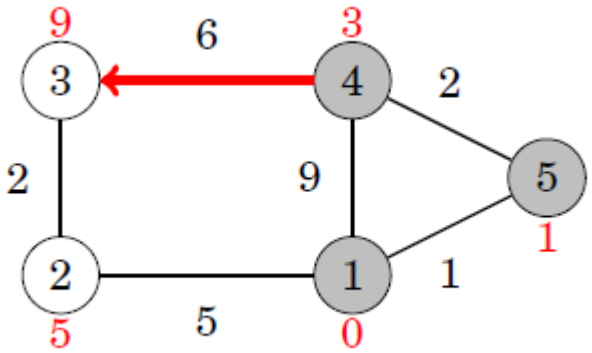


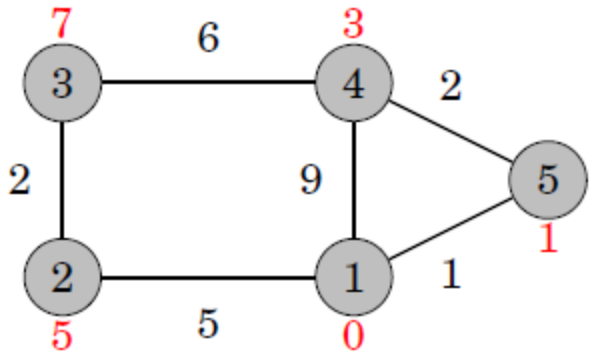
Dijkstra's Algorithm O(n^2) veya O(n+m logm)

Floyd Warshall Algorithm
O(n^3) tür ancak kodlaması kolaydır.


∞
6
9
3
4
2
2
9
5
1
2
1
1
5
5
0


∞
6
3
3
4
2
2
9
5
1
2
1
1
5
5
0


9
6
3
3
4
2
2
9
5
1
2
1
1
5
5
0


7
6
3
3
4
2
2
9
5
1
2
1
1
5
5
0

```
for (int i = 1; i <= n; i++) distance[i] = INF;
distance[x] = 0;
q.push({0,x});
while (!q.empty()) {
    int a = q.top().second; q.pop();
    if (processed[a]) continue;
    processed[a] = true;
    for (auto u : adj[a]) {
        int b = u.first, w = u.second;
        if (distance[a]+w < distance[b]) {
            distance[b] = distance[a]+w;
            q.push({-distance[b],b});
        }
    }
}
```

Kodu incele

İnceleme soruları:

UVA 10986: Sending e-mail
UVA 11377: Airport setup